

=== GNB (1780946787) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.

This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.

To run the srsRAN Project gNB with this config, use the following command:

sudo ./gnb -c gnb_zmq.yaml

cu_cp:

amf:
 addr: 10.53.1.2 # The address or hostname of the AMF.
 port: 38412
 bind_addr: 10.53.1.1 # A local IP that the gNB binds to for traffic from the AMF.
 supported_tracking_areas:
 - tac: 7
 plmn_list:
 - plmn: "00101"
 tai_slice_support_list:
 - sst: 1 # 3GPP TS 23.501 Table 5.15.2-1: SST=1 for eMBB
 inactivity_timer: 2700 # Sets the UE/PDU Session/DRB inactivity timer to 2700 seconds (45 minutes). Supported: [1 - 7200].

ru_sdr:

 device_driver: zmq # The RF driver name.
 device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass arguments to the selected RF driver.
 srate: 11.52 # RF sample rate might need to be adjusted according to selected bandwidth.
 tx_gain: 75 # Transmit gain of the RF might need to adjusted to the given situation.
 rx_gain: 75 # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:

 dl_arfcn: 368500 # ARFCN of the downlink carrier (center frequency).
 band: 3 # The NR band.
 channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.
 common_scs: "15kHz" # Subcarrier spacing in kHz used for data.
 ssb_scs: "15kHz" # SSB subcarrier spacing in kHz. Must match common_scs.
 plmn: "00101" # PLMN broadcasted by the gNB.
 tac: 7 # Tracking area code (needs to match the core configuration).
 pdcch:
 common:
 ss0_index: 0 # Set search space zero index to match srsUE capabilities
 coreset0_index: 6 # Set search CORESET Zero index to match srsUE capabilities
 dedicated:
 ss2_type: common # Search Space type, has to be set to common
 dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
 prach:
 prach_config_index: 1 # Sets PRACH config to match what is expected by srsUE
 pdsch:
 mcs_table: qam64 # Sets PDSCH MCS to 64 QAM
 pusch:
 mcs_table: qam64 # Sets PUSCH MCS to 64 QAM

log:

 filename: /tmp/gnb.log # Path of the log file.
 all_level: info # Logging level applied to all layers.
 hex_max_size: 0

pcap:

 mac_enable: false # Set to true to enable MAC-layer PCAPs.
 mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
 ngap_enable: true # Set to true to enable NGAP PCAPs.
 ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
 e2ap_enable: true # Set to true to enable E2AP PCAPs.
 e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
 e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.

e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:

```
# enable_du_e2: true          # Enable DU E2 agent (one for each DU instance)
# e2sm_kpm_enabled: true      # Enable KPM service module
# e2sm_rc_enabled: true       # Enable RC service module
# addr: 127.0.0.1             # RIC IP address
# bind_addr: 127.0.0.100      # A local IP that the E2 agent binds to for traffic from the RIC. ONLY
                              # required if running the RIC on a separate machine.
# port: 36421                 # RIC port
```

metrics:

layers:

```
    enable_rlc: true          # Enable RLC metrics
    enable_sched: true        # Enable DU scheduler metrics
periodicity:
    du_report_period: 400     # Set DU statistics report period to 400ms
```

=== UE (1780946787) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780946787) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#
```

services:

5gc:

container_name: open5gs_5gc

build:

context: open5gs

target: open5gs

args:

OS_VERSION: "22.04"

OPEN5GS_VERSION: "v2.7.0"

environment:

- MONGODB_IP=127.0.0.1

- OPEN5GS_IP=10.53.1.2

- UE_IP_BASE=10.45.0

- UPF_ADVERTISE_IP=10.53.1.2

- DEBUG=false

SUBSCRIBER_DB=001010000000001,00aabbccddeeff112233445566778899,opc,112233445566778899aabbccddeeff00,8000,9,10.4
5.1.2

- NETWORK_NAME_FULL=srsRAN

- NETWORK_NAME_SHORT=srsRAN

- TZ=Europe/Madrid

- T3512_VALUE=2700

privileged: true

ports:

- "9898:9999/tcp"

Uncomment port to use the 5gc from outside the docker network

- "38412:38412/sctp"

- "2152:2152/udp"

command: 5gc -c open5gs-5gc.yml

healthcheck:

test: ["CMD-SHELL", "nc -z 127.0.0.20 7777"]

interval: 3s

timeout: 1s

retries: 60

networks:

ran:

ipv4_address: \${OPEN5GS_IP:-10.53.1.2}

gnb:

container_name: srsran_gnb

Build info

image: srsran/gnb

build:

context: ..

```

    dockerfile: docker/Dockerfile
    args:
        OS_VERSION: "24.04"
environment:
    - GNB_IP=10.53.1.1
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
volumes:
    # Access USB to use some SDRs
    - /dev/bus/usb/:/dev/bus/usb/
    # Sharing images between the host and the pod.
    # It's also possible to download the images inside the pod
    - /usr/share/uhd/images:/usr/share/uhd/images
    # Save logs and more into gnb-storage
    - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
    - gnb_config.yml
    - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
    ran:
        ipv4_address: ${GNB_IP:-10.53.1.1}
    metrics:
        ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
    5gc:
        condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
    file: ${GNB_CONFIG_PATH:-../configs/gnb_rf_b200_tdd_n78_20mhz.yml} # Path to your desired config file
gnb_compose_config.yml:
    content: |
        cu_cp:
            amf:
                addr: ${OPEN5GS_IP:-10.53.1.2}
                bind_addr: 0.0.0.0
        metrics:
            autostart_stdout_metrics: true
            enable_json: true
        remote_control:
            bind_addr: 0.0.0.0
            enabled: true

volumes:
    gnb-storage:

networks:
    ran:
        ipam:
            driver: default
            config:
                - subnet: 10.53.1.0/24
    metrics:
        ipam:
            driver: default

```

config:

- subnet: 172.19.1.0/24

=== NOTAS (1780946787) ===

- PLMN (MCC 001, MNC 01, TAC 7) consistente en gNB, UE y Core (Regla 1).
- dl_arfcn 368500 para Banda 3 es coherente (Regla 2).
- common_scs y ssb_scs configurados como "15kHz" y posicionados correctamente (Regla 3).
- Puertos ZMQ cruzados entre gNB (tx=2000, rx=2001) y UE (tx=2001, rx=2000) (Regla 4).
- bind_addr configurado según la Regla 5 (10.53.1.1 en gnb_zmq.yaml, 0.0.0.0 en docker-compose).
- Bloque ru_sdr copiado inmutablemente del CAG, srate 11.52 MHz compatible con 10 MHz de ancho de banda (Regla 6).
- IMSI, K y OPC consistentes y en el formato correcto en UE y SUBSCRIBER_DB del Core (Regla 7).
- SST para eMBB (1) añadido con comentario de la fuente RAG.
- Inactivity timer (duración de sesión) configurado a 2700 segundos (45 minutos) en gNB y T3512_VALUE en 5GC.
- Métricas RLC y Scheduler DU habilitadas con periodo de reporte de 400 ms.
- Capturas PCAP de NGAP habilitadas en gNB.